

GET 208: STRENGTH OF MATERIALS

STRESSES AND STRAINS DUE TO LOADING

The strength of a material depends on its ability to sustain a load without undue deformation or failure. One of the most important tests to perform in order to determine this property is the tension or compression test. Although several important mechanical properties of a material can be determined from this test, it is used primarily to determine the relationship between the average normal stress and average normal strain in many engineering materials such as metals, ceramics, polymers, and composites. The load and corresponding deformation data are used to calculate various values of the stress and corresponding strain in the specimen. A plot of the results produces a curve called the stress–strain diagram. A typical stress–strain diagram is shown in Figure 1 below.

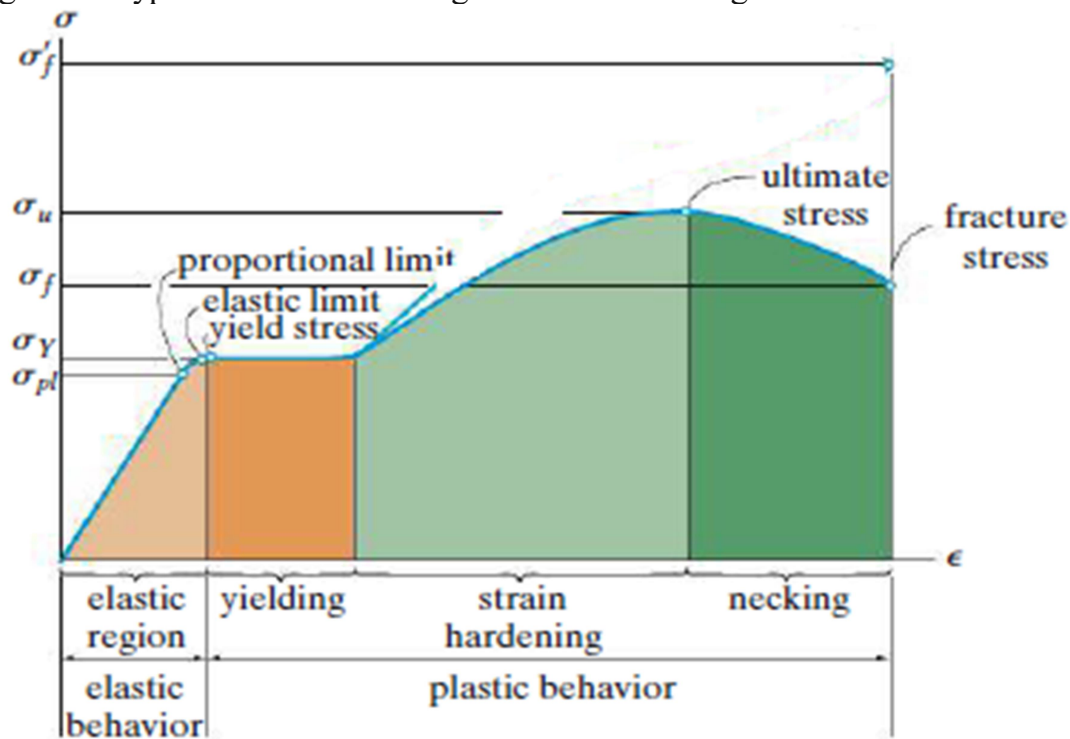


Figure 1: Conventional Stress-Strain Diagram

Elastic Behavior: Elastic behavior of the material occurs when the strains in the specimen are within the elastic region as shown in Figure 1. The curve is actually a *straight line* throughout most of the region, so that the stress is *proportional* to the strain. The material in this region is said to be *linear elastic*. The upper stress limit to this linear relationship is called the *proportional limit*, σ_{pl} . If the stress slightly exceeds the proportional limit, the curve tends to bend and flatten out as shown. This continues until the stress reaches the *elastic limit*. Upon reaching this point, if the load is removed the specimen will still return back to its original shape.

Yielding: A slight increase in stress above the elastic limit will cause the material to *deform permanently*. This behavior is called *yielding*. The stress that causes yielding is called the *yield stress*, σ_Y , and the deformation that occurs is called *plastic deformation*.

Strain Hardening: When yielding has ended, an increase in load can be supported by the specimen, resulting in a curve that rises continuously but becomes flatter until it reaches a maximum stress referred to as the *ultimate stress*, σ_u . The rise in the curve in this manner is called *strain hardening*.

Necking: Up to the ultimate stress, as the specimen elongates, its cross-sectional area will decrease. This decrease is fairly *uniform* over the specimen's entire length; however, just after, at the ultimate stress, the cross-sectional area will begin to decrease in a *localized* region of the specimen. As a result, a “neck” tends to form in this region as the specimen elongates further. Here the stress–strain diagram tends to curve downward until the specimen breaks at the *fracture stress*, σ_f .

Strain Energy: As a material is deformed by an external loading, it tends to store energy internally throughout its volume. Since this energy is related to the strains in the material, it is referred to as *strain energy*. Strain energy is denoted as U . For applications, it is often convenient to specify the strain energy per unit volume (V) of material. This is called the *strain-energy density*, and it can be expressed as:

$$u = \frac{\Delta U}{\Delta V} = \frac{1}{2} \sigma \epsilon$$

If the material behavior is *linear elastic*, then Hooke's law applies, $\sigma = E\epsilon$, and therefore the above equation can be expressed as:

$$u = \frac{1}{2} \frac{\sigma^2}{E}$$

Modulus of Resilience: When the stress reaches the proportional limit, the strain-energy density is referred to as the *modulus of resilience*, i.e.

$$u_r = \frac{1}{2} \sigma_{pl} \epsilon_{pl} = \frac{1}{2} \frac{\sigma_{pl}^2}{E}$$

GENERALIZED HOOKE'S LAW

The stress-strain diagrams for most engineering materials exhibit a linear relationship between stress and strain within elastic limit. Consequently, an increase in stress causes a proportionate increase in strain. This fact is known as Hooke's law. It is expressed as $\sigma = E\epsilon$.

E represents the constant of proportionality, which is called the modulus of elasticity or Young's modulus. The equation above represents the equation of the initial straight-lined portion of the stress-strain diagram up to the proportional limit. Furthermore, the modulus of elasticity represents the slope of this line. Also, the modulus of elasticity is a mechanical property that indicates the stiffness of a material. Materials that are very stiff, such as steel, have large values of E ($E_{st} = 200GPa$), whereas soft materials such as vulcanized rubber may have low values ($E_r = 0.7MPa$).

A similar modulus called the shear modulus or the modulus of rigidity relates the shearing stress τ and the shearing strain γ as follows:

$$G = \frac{\tau}{\gamma}$$

Poisson's ratio: A material loaded in one direction will undergo strains perpendicular to the direction of the load in addition to those parallel to the load. The ratio of the lateral (perpendicular) strain to the longitudinal (axial) strain is called Poisson's ratio and is given as:

$$\nu = -\frac{\epsilon_{lateral}}{\epsilon_{longitudinal}}$$

Consider a bar having an original radius r and length L and subjected to the tensile force P . This force elongates the bar by an amount δ and its radius contracts by an amount δ' . Strains in the longitudinal or axial direction and in the lateral or radial direction are given as follows:

$$\epsilon_{longitudinal} = \frac{\delta}{L} \quad \text{and} \quad \epsilon_{lateral} = \frac{\delta'}{r}$$

Like the modulus of elasticity and the shear modulus G , Poisson's ratio is also a property of a material. Hence, Poisson's ratio (ν) is related to E and G by the formula:

$$E = 2(1 + \nu)G$$

EXERCISES

1) An aluminum specimen has a diameter of 25mm and a length of 250mm. If a force of 165kN elongates the length by 1.2mm, determine the modulus of elasticity. Also determine by how much the force causes the diameter of the specimen to contract. Take $G_{al} = 26GPa$ and $\sigma_Y = 440 MPa$. **(Ans: $E = 70GPa$ & $\delta' = 0.0416mm$)**

2) A square steel rod 15mmx15mm in section is to carry an axial load (compressive) of 150 kN. Calculate the shortening in a length of 75mm. $E = 2.03 \times 10^8 kN/m^2$. **(Ans: $\delta = 0.246mm$)**

3) The following observations were made during a tensile test on a mild steel specimen 50mm in diameter and 180mm long. Elongation with 50kN load (within limit of proportionality) is $\delta = 0.256mm$, yield load = 150kN, Maximum load = 230 kN, length of specimen at fracture = 232mm. Determine (i) Modulus of Elasticity (ii) Yield point stress (iii) Ultimate Stress (iv) Percentage elongation.

(Ans: (i) $E = 17.96GPa$ (ii) $\sigma_Y = 76.4MPa$,

(iii) $\sigma_U = 117.14MPa$, (iv) $\delta_{percentage} = 28.9\%$)

4) A hollow cast-iron cylinder 5m long, 400mm outer diameter and metal thickness of 50mm is subjected to a central load on the top when standing straight. The stress produced is $80,000 kN/m^2$. Assume Young's modulus for cast iron is $1.5 \times 10^8 kN/m^2$. Determine (i) Magnitude of the load (ii) longitudinal strain produced (iii) Total decrease in length.

(Ans: (i) $P = 4398.2kN$ (ii) $\epsilon_{long} = 5.33 \times 10^{-4}$, (iii) $\delta = 2.67mm$)

STRESSES AND STRAINS DUE TO TEMPERATURE CHANGES

If the temperature of a body is lowered or raised, its dimensions will decrease or increase correspondingly. If these changes are checked, the stresses thus developed in the body are called temperature stresses and the corresponding strains are called temperature strains.

If L = length of a bar of uniform cross-section

t_1 = Initial temperature of the bar

t_2 = Final temperature of the bar

α = Coefficient of linear expansion

The extension (e) in the bar due to rise in temperature will be:

$$e = \alpha(t_2 - t_1)L$$

If this elongation in the bar is prevented by some external force or by fixing the bar ends, the temperature strain thus produced will be given by:

$$\text{Temperature strain} = \frac{\alpha(t_2 - t_1)L}{L} = \alpha(t_2 - t_1)$$

Therefore, temperature stress developed = $\alpha(t_2 - t_1)E$

Exercises

1) A steel rod 15m long is at a temperature of 15°C. Find the free extension of the length when the temperature is raised to 60°C. Determine the temperature stress produced when:

- i. The expansion of the rod is prevented.
- ii. The rod is permitted to expand by 6mm. Take $\alpha = 12 \times 10^{-6}$ per °C and $E = 200$ GPa

2) Calculate the values of stress and strain in a non-prismatic steel bar ABC which is rigidly restrained at its ends A and C. The length of both AB and BC is 0.3m and the cross-sectional areas of AB and BC are 400mm² and 800mm², respectively. A close fit exists at both of the rigid supports at room temperature and the temperature is raised by 75 °C. Take $E = 200$ GPa and $\alpha = 12 \times 10^{-6}$ per °C for steel.

References:

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